# Two-Arc Integer Lipschitz Heights: Boundary-Displacement Energy and Certified Interior Gaussian Comparisons

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M444

October 9, 2026

**Abstract**

We analyze the two-arc triangular-lattice height manuscript with its odd-height, physical-area and inverse-Laplacian conventions retained. The source's Gaussian field identification, conditional cut comparisons, boundary arches and angular normalization sum remain upstream inputs, not conclusions of a finite certificate. Our distinct local contribution is an exact minimum Dirichlet-energy formula for comparing two boundary prescriptions on a buffered conformal disk. It yields exact relative entropy and total variation for the corresponding proposed Gaussian laws, positive Fourier and product series with explicit tails, and different sharp asymptotics for small mark motion and vanishing observation buffers. This resolves a concrete trace obstruction: a two-arc step mean is bounded harmonic but is not generally an $H^1$ lift. A secondary conditional certificate propagates finite counting, Fourier quadrature and a separately supplied angular-tail budget to the source's stiffness and the new comparison bounds. No effective lattice rate, interface limit or independent proof of the global field theorem is claimed. Contribution assessment: C3/5, a source-specific buffered boundary-displacement certificate with exact series, sharp limit regimes and explicit normalization uncertainty, conditional on the inherited field identification.

## Contents

**1 The source model and the boundary problem** **2**

**2 Step data are not global finite-energy lifts** **3**

**3 The exact buffered extension energy** **4**

**4 Exact comparison of the proposed Gaussian laws** **4**

**5 Positive series, certified tails and mark motion** **6**

**6 The existing normalization series and a conditional certificate** **8**

**7 Source boundaries, literature and consequences** **9**

**8 Verification and contribution boundary** **10**

# 1 The source model and the boundary problem

Let $\mathbb{T} = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. The source [1] uses simply connected triangular polygon approximations $D_\delta$ of a bounded smooth simply connected domain $D$, with uniformly convergent marked boundary parametrizations. Two distinct edge-midpoint marks separate the boundary vertices into disjoint arcs. The uniform law is on odd integer heights $h_\delta$, with adjacent differences in $\{0, 2, -2\}$, and values $+1, -1$ on the two arcs. Heights are interpolated affinely on each triangle and paired with tests using Euclidean area. This is not the weighted zero-boundary model of M443, nor a real-valued uniformly convex gradient model.

Put $H_\delta = (h_\delta - 1)/2$. Its boundary values are $0, -1$, and its adjacent differences have absolute value at most one. The source claims

$$H_\delta \Rightarrow m + \sqrt{v}\,\Phi_D, \qquad h_\delta \Rightarrow \bar{g} + \sigma\Phi_D, \qquad \bar{g} = 1 + 2m, \quad v = 4\pi^2 c_*, \quad \sigma^2 = 16\pi^2 c_*. \tag{1.1}$$

The field $\Phi_D$ has Dirichlet covariance $G_D = (-\Delta_D)^{-1}$. Its logarithmic singularity is $\frac{1}{2\pi}\log(1/|x-y|)$. The means are bounded harmonic extensions of their respective step data. The asserted topology includes $H^{-t}_{\mathrm{loc}}(D)$, $t > 1$, with all finite joint test moments. Centering subtracts the *actual* discrete expectation, whose convergence is a separate source step.

The exact local spin encoding is

$$B = +1 \text{ on } H \equiv 0, 1 \pmod 4, \quad W = +1 \text{ on } H \equiv -1, 0 \pmod 4, \quad dH(x,y) = \frac{W_x B_y - B_x W_y}{2}. \tag{1.2}$$

Coherence excludes simultaneous changes of both colors. On each triangular face one color is constant, hence the increment has zero curl. Simple connectivity then gives a height lift once one anchor is fixed. This finite identity does not by itself give a Gaussian field.

## 1.1 The full proof architecture

The following map refers to the actual scientific source files, all freshly read for this report. Reading the whole manuscript does not amount to independently proving every theorem it uses.

| Source section | Mechanism and retained obligation |
|---|---|
| 02-spin | Gray lift, equality-constrained spin FKG, exact one-color cuts; selected cuts must leave their strict interior unexamined. |
| 03-cuts | Imported loop bounds, buffered circuits, plane mixing and multiplicative probabilities of separated rare pins. |
| 04-moments | Rounded midpoint tails (already exponential in the source), loop smearing bounds, fair component switches and offset parity classes. |
| 05-spectral | Positive normal transfer with auxiliary phases, reciprocal aliases, Cauchy mixtures, dispersion defect and signed angular sum. |
| 06-normalization | Absolutely convergent finite-volume double series; positive $c_*$, odd-height factor two, physical-area cancellation. |
| 07-pins | Actual integer brackets, normalized rare-pin reflection norms, analytic translations, exact Fourier projection onto equal offsets. |
| 08-boundary | Attachments and arches under actual usable traces; stopped conditional Gibbs regions, not synthetic laws used as actual ones. |
| 09-limits | Tightness, bounded harmonic boundary mean, deterministic many-point references, absolute moments and removal of diagonal terms. |
| 10-gaussian | Conditional plane matching independent of the entire exterior record, collision coefficients, Gaussian recursion and determinacy. |

The source first fixes positive geometric clearances and takes the mesh limit; gaps, windows and finite grids are subsequently sent through their specified limits. Boundary references take depth to zero at a fixed number of distinct points, then increase that number. None of these statements supplies an effective joint rate for degenerating geometry.

### 1.2 What the present report adds

We compare *two different marked-arc prescriptions* for the proposed limit, in the same domain and at the same true $\sigma > 0$. The central result is a scalar Fourier evaluation of the full minimum extension energy, not a finite covariance matching claim. It retains every angular mode and then supplies an upper bound on the discarded tail. Classical harmonic analysis and Gaussian translation are credited as such. Their assembly for this two-arc normalization and the sharp boundary-motion regimes is the local increment assessed here. The M443 rare-event modulus and conditional Green residual results are not reused as new contributions.

## 2 Step data are not global finite-energy lifts

In the unit disk write $I = (\alpha, \beta)$, with a chosen angular lift $0 < \beta - \alpha < 2\pi$, for the plus arc, and put $b_I = 2\mathbf{1}_I - 1$. With $c_n = (2\pi)^{-1} \int_0^{2\pi} b_I(\theta) e^{-in\theta}\, d\theta$, direct integration gives

$$c_0 = \frac{\beta - \alpha}{\pi} - 1, \qquad c_n = \frac{e^{-in\alpha} - e^{-in\beta}}{\pi i n} \quad (n \neq 0), \qquad \bar{g}_I(re^{i\theta}) = \sum_{n \in \mathbb{Z}} c_n r^{|n|} e^{in\theta}. \tag{2.1}$$

The series converges smoothly on every interior compact. The bounded Poisson extension is unique; values assigned to the two endpoints do not affect it.

**Proposition 2.1** (Trace obstruction and buffered deterministic bound)**.** *The step data $b_I$ are not in $H^{1/2}(\partial\mathbb{D})$, and $\bar{g}_I \notin H^1(\mathbb{D})$. If two oriented arcs have endpoint movements with total angular length at most $s$, without wrapping the chosen lifts, then for $|z| \leq \rho < 1$,*

$$|\bar{g}_J(z) - \bar{g}_I(z)| \leq \frac{1+\rho}{\pi(1-\rho)} s. \tag{2.2}$$

*Near a straight boundary jump of size $J$, the angular singular lift has energy $J^2 \pi^{-1} \log(R/\varepsilon)$ in a half-annulus of radii $\varepsilon, R$.*

*Proof.* The Fourier criterion for $H^{1/2}$ requires $\sum_{n \geq 1} n|c_n|^2 < \infty$. Here $n|c_n|^2 = 2[1 - \cos(n(\beta - \alpha))]/(\pi^2 n)$. The harmonic series diverges, whereas the cosine series converges by summation by parts, since its partial sums are bounded for a nontrivial arc. The harmonic energy on a disk of radius $r$ is $4\pi \sum_{n \geq 1} n|c_n|^2 r^{2n}$; monotone convergence proves its divergence. This also prevents any $H^1$ lift with this trace.

The Poisson kernel $(1-|z|^2)/|e^{i\theta} - z|^2$ is at most $(1+\rho)/(1-\rho)$. The boundary differences have magnitude at most two and support of length at most $s$, giving (2.2). In polar coordinates about a straight jump, the lift is a constant plus $J\theta/\pi$ on $0 < \theta < \pi$. Its squared gradient is $J^2/(\pi^2 r^2)$, whose integral is the stated logarithm. □

The freshly read original TR8 proof [6] formulates the coercive inverse on $H_0^1$, with boundary liftings only for $H^{1/2}$ traces. Its Green-equivalence proof subtracts that lifting before inversion; it does not cover a step trace by silently declaring it $H^1$. The original TR250 [7] proves physical-energy orthogonality and emphasizes conformity and actual boundary conditions. We use precisely that minimum-energy logic below, on a *buffered matching problem*. No claim of a global energy residual for the unmodified step mean is made.

## 3 The exact buffered extension energy

Let $B \in L^2(\partial\mathbb{D};\mathbb{R})$, let $d_n$ be its Fourier coefficients in the convention of (2.1), and let $w = P[B]$ be its Poisson extension. For $0 < \rho < 1$ set $T = \log(1/\rho) > 0$. Define

$$E_\rho(B) = \inf\left\{\int_{\mathbb{D}} |\nabla u|^2\, dx : u \in H_0^1(\mathbb{D};\mathbb{R}),\ u = w \text{ a.e. on } \rho\mathbb{D}\right\}. \tag{3.1}$$

The matching set is nonempty: multiply the smooth interior harmonic function by a cutoff equal to one on a slightly larger disk. It is a closed affine subset of the energy Hilbert space, so has a unique minimizer.

**Theorem 3.1** (Boundary-to-interior energy formula)**.** *For every real $B \in L^2(\partial\mathbb{D})$,*

$$E_\rho(B) = \frac{2\pi d_0^2}{T} + 8\pi \sum_{n=1}^{\infty} \frac{n|d_n|^2}{e^{2nT}-1}. \tag{3.2}$$

*The minimizer is $w$ inside the disk and, on $\rho < r < 1$, is*

$$u_*(re^{i\theta}) = d_0 \frac{\log r}{\log \rho} + \sum_{n\neq 0} d_n \frac{\rho^{2|n|}(r^{-|n|} - r^{|n|})}{1-\rho^{2|n|}} e^{in\theta}. \tag{3.3}$$

*In particular, the energy is finite for every fixed buffer, even for step data with no global $H^1$ harmonic lift.*

*Proof.* First take a real trigonometric polynomial. The displayed annular function is harmonic, matches $w$ at $r = \rho$, and vanishes at $r = 1$. It therefore gives a conforming $H_0^1$ function. Every admissible perturbation vanishes in $\rho\mathbb{D}$ and has zero trace on both annular boundaries. Integration by parts there gives orthogonality to $u_*$. Consequently its energy adds to $\|u_*\|_\nabla^2$, proving minimality.

For a real mode $ar^n\cos(n\theta)$, the inner energy is $\pi n a^2 \rho^{2n}$. The outer energy, by its inward boundary flux, is $\pi n a^2 \rho^{2n}(1+\rho^{2n})/(1-\rho^{2n})$. Their sum is $2\pi n a^2 \rho^{2n}/(1-\rho^{2n})$. The sine mode gives the same expression, and angular orthogonality removes cross terms. For the complex convention, the squared real cosine and sine coefficients add to $4|d_n|^2$. The constant annular mode has energy $2\pi d_0^2/T$. These computations give (3.2).

For general $B$, Parseval gives square-summable $d_n$. The weights $n/(e^{2nT}-1)$ are bounded for fixed $T > 0$, so polynomial minimizers converge in $H_0^1$. Their interior restrictions converge to the Poisson extension, and their orthogonality persists. This proves the formula, the minimizer and convergence of its series in energy. □

*Remark* 3.2 (A distinction between two energies). The energy of $w$ on $\rho\mathbb{D}$ alone is not $E_\rho$. The annular extension is required because the covariance is the restriction of $G_{\mathbb{D}}$, not the Green kernel of a new zero-boundary GFF on $\rho\mathbb{D}$. Discarding the annular energy would give an incorrect comparison, even for a constant mean shift, whose inner energy is zero.

## 4 Exact comparison of the proposed Gaussian laws

Let $\mu_I^\rho$ be the law of $(\bar{g}_I + \sigma\Phi_{\mathbb{D}})|_{\rho\mathbb{D}}$ as a random distribution, with one fixed $\sigma > 0$. These are well-defined Gaussian laws independently of the lattice source. Applying the result to limits of Lipschitz heights is conditional on the source identification (1.1). Write $\Phi_{\mathrm{N}}$ for the standard normal distribution function, and use $\mathrm{TV}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)|$.

**Theorem 4.1** (Exact local distinguishability)**.** *For $B = b_J - b_I$, put $d = \sqrt{E_\rho(B)}/\sigma$. Then*

$$\mathrm{KL}(\mu_J^\rho \| \mu_I^\rho) = \frac{d^2}{2}, \qquad \mathrm{TV}(\mu_J^\rho, \mu_I^\rho) = 2\Phi_{\mathrm{N}}(d/2) - 1. \tag{4.1}$$

*The laws are mutually absolutely continuous for every $\rho < 1$. The same assertions hold for any two real $L^2$ boundary data with harmonic means and the same covariance $\sigma^2 G_{\mathbb{D}}$.*

*Proof.* Here is the Gaussian translation argument, including what is observable on the smaller disk. Let $\mathcal{H} = H_0^1(\mathbb{D}; \mathbb{R})$, with inner product $a(u, v) = \int \nabla u \cdot \nabla v$. Realize the GFF through a linear Gaussian family $W(u)$, $u \in \mathcal{H}$, of covariance $a(u, v)$, so $\Phi_{\mathbb{D}}(f) = W(G_{\mathbb{D}} f)$. This is the classical Gaussian Hilbert-space construction [5]. Let

$$\mathcal{M}_\rho = \overline{\mathrm{span}\{G_{\mathbb{D}} f : f \in C_c^\infty(\rho\mathbb{D}; \mathbb{R})\}}^{\mathcal{H}}.$$

Its orthogonal complement consists precisely of the $\mathcal{H}$ functions vanishing a.e. on $\rho\mathbb{D}$, because $a(u, G_{\mathbb{D}} f) = \int uf$. The minimizer in (3.1) belongs to $\mathcal{M}_\rho$ by its variational orthogonality. Thus $Z = W(u_*)/\sigma$ is measurable from the restricted field, as an $L^2$ limit of finite linear test statistics. It is normal with variance $d^2$. The random variable

$$L = \exp(Z - d^2/2)$$

has expectation one. Tilting by $L$ shifts the mean of $\sigma W(G_{\mathbb{D}} f)$ by $a(u_*, G_{\mathbb{D}} f) = \int wf$, with covariance unchanged. This follows by the joint finite-dimensional Gaussian moment generating function; a countable determining set of tests identifies the law. Hence $L$ is the likelihood ratio for the two restricted laws. Under the shifted law, $Z$ has mean $d^2$ and variance $d^2$, so $\mathbb{E}_J \log L = d^2/2$. The maximizing set for total variation is $\{L \geq 1\} = \{Z \geq d^2/2\}$. The difference of its two Gaussian probabilities is $2\Phi_{\mathrm{N}}(d/2) - 1$. The case $d = 0$ follows directly. Positivity of both likelihoods gives equivalence. □

**Corollary 4.2** (Conformal domains, observations and coupling)**.** *Let $F : \mathbb{D} \to D$ be a conformal map extending to the boundary of a smooth Jordan domain. Transfer the two arc prescriptions by this map. On $U_\rho = F(\rho\mathbb{D})$, formulas (3.2)–(4.1) hold with the angular coefficients of their pullbacks. A smaller observation region has no larger entropy or total variation. For a finite test vector with covariance $\sigma^2 C$, $C$ positive definite, and mean difference $\Delta$,*

$$\Delta^T C^{-1} \Delta \leq E_\rho(B), \qquad \mathrm{KL}_{\mathrm{tests}} = \frac{\Delta^T C^{-1} \Delta}{2\sigma^2}. \tag{4.2}$$

*Using the same GFF for the two laws gives a coupling whose difference is exactly the deterministic harmonic function $w$.*

*Proof.* Conformal change of variables preserves Dirichlet energy in dimension two: the squared derivative factor cancels the area Jacobian. It maps zero-trace energy spaces and the matching constraints bijectively, preserving the infimum. Test densities transform with their area Jacobian; treating them as scalars without this factor would be incorrect. This also preserves the GFF covariance by the weak inverse identity. Restriction of the likelihood ratio is conditional expectation, so convexity of $x \log x$ and the total-variation definition give contraction. For finite tests, the squared energy of the projection of $u_*$ onto their Green potentials is $\Delta^T C^{-1} \Delta$; Pythagoras proves the bound. The coupling statement follows by subtracting the same Gaussian field. □

These comparisons concern a *common true variance*. Changing the variance multiplier of an infinite-dimensional GFF is not a small total-variation perturbation merely because the scalars are close. The scalar uncertainty treated below is an enclosure of the same unknown true $\sigma$, not a comparison of two artificially rescaled full fields. Nor does weak convergence of the lattice fields imply convergence in entropy or total variation. For any one fixed bounded continuous functional of finitely many interior tests, the source convergence would pass its expectation to the limits compared here; no supremum over mesh-dependent events or simultaneous moving-mark lattice rate is obtained.

## 5 Positive series, certified tails and mark motion

Write $q = \rho^2$. Every term of (3.2) is nonnegative. Let $E_{\rho,K}$ be its constant term plus modes $1 \le n \le K$.

**Proposition 5.1** (A two-arc finite spectral certificate)**.** *For the difference of any two plus-arc prescriptions,*

$$0 \le E_\rho - E_{\rho,K} \le U_{\rho,K} := \frac{128}{\pi} \frac{q^{K+1}}{(K+1)(1-q)(1-q^{K+1})}, \qquad K \ge 0. \tag{5.1}$$

*Consequently both quantities in (4.1) have rigorous lower and upper bounds obtained by substituting $E_{\rho,K}$ and $E_{\rho,K} + U_{\rho,K}$. Validated coefficient or arithmetic errors must be included separately.*

*Proof.* Equation (2.1) gives $|d_n| \le 4/(\pi n)$. The tail is therefore at most $(128/\pi)\sum_{n>K} q^n/[n(1-q^n)]$. Replace $n$ by $K+1$ and $1-q^n$ by $1-q^{K+1}$ in the denominators, and sum the geometric series. Monotonicity of the entropy and normal distribution functions gives the enclosures. This is an upper bound on the *whole* omitted angular complement, not an inference from a finite positive Gram matrix. □

**Theorem 5.2** (One moving endpoint: exact product and two sharp regimes)**.** *Fix one endpoint $\alpha$, and replace $\beta$ by $\beta + \delta$, where $0 < \delta < 2\pi - (\beta - \alpha)$. The boundary difference is two on the added arc of length $\delta$ and zero elsewhere. For $T > 0$,*

$$E_\rho(\delta) = \frac{2\delta^2}{\pi T} + \frac{16}{\pi} \sum_{n \ge 1} \frac{1 - \cos(n\delta)}{n(e^{2nT} - 1)} \tag{5.2}$$

$$= \frac{2\delta^2}{\pi T} + \frac{8}{\pi} \sum_{k \ge 1} \log\left(1 + \frac{\sin^2(\delta/2)}{\sinh^2(kT)}\right). \tag{5.3}$$

$$\lim_{\delta \downarrow 0} \frac{E_\rho(\delta)}{\delta^2} = J(T) := \frac{2}{\pi T} + \frac{2}{\pi} \sum_{k \ge 1} \frac{1}{\sinh^2(kT)}, \qquad \lim_{T \downarrow 0} T^2 J(T) = \frac{\pi}{3}. \tag{5.4}$$

*More generally, for every fixed real $B \in L^2(\partial\mathbb{D})$,*

$$\lim_{T \downarrow 0} T E_{e^{-T}}(B) = \int_0^{2\pi} |B(\theta)|^2 \, d\theta. \tag{5.5}$$

*In particular, at a fixed nonzero displacement, $E_{e^{-T}}(\delta) \sim 4\delta/T$. The small-displacement and thin-buffer asymptotics are different ordered limits, not a uniform joint expansion.*

*Proof.* Direct integration gives $d_0 = \delta/\pi$ and $|d_n|^2 = 2[1-\cos(n\delta)]/(\pi^2 n^2)$, proving (5.2). Expand $(e^{2nT}-1)^{-1} = \sum_{k\geq 1} q^{kn}$. All summands after multiplication by $1-\cos(n\delta)$ are nonnegative, so Tonelli applies. For $0 < a < 1$, the absolutely convergent power series for the logarithm gives

$$\sum_{n\geq 1} \frac{a^n(1-\cos(n\delta))}{n} = \frac{1}{2}\log\frac{1-2a\cos\delta+a^2}{(1-a)^2}.$$

Taking $a = q^k$ and using $4q^k/(1-q^k)^2 = 1/\sinh^2(kT)$ gives (5.3). For fixed $T$, $\log(1+x) \leq x$ and $\sin^2(\delta/2) \leq \delta^2/4$ supply a summable dominating sequence after division by $\delta^2$. This proves the first limit in (5.4). For the second, $T^2/\sinh^2(kT) \leq 1/k^2$ and the pointwise limit is $1/k^2$. Dominated convergence and the classical identity $\sum k^{-2} = \pi^2/6$ yield $\pi/3$.

Finally multiply (3.2) by $T$. For each positive mode, $2nT/(e^{2nT}-1) \to 1$ and lies in $[0,1]$. Parseval and dominated convergence give $2\pi d_0^2 + 4\pi\sum_{n\geq 1}|d_n|^2 = \|B\|_2^2$. The single added arc has squared boundary norm $4\delta$. □

**Corollary 5.3** (A second positive truncation bound)**.** *Truncating the product series* (5.3) *at* $k = K$ *gives a lower bound, with remainder at most*

$$\frac{32}{\pi}\sin^2(\delta/2)\frac{q^{K+1}}{(1-q)(1-q^{K+1})^2}. \tag{5.6}$$

*Proof.* Bound each logarithm by its argument and use $\sinh^{-2}(kT) = 4q^k/(1-q^k)^2$. Replace the denominator by $(1-q^{K+1})^2$ and sum the resulting geometric tail. □

The positive product is useful when cancellation in $1-\cos(n\delta)$ would be inconvenient; use $2\sin^2(n\delta/2)$ in actual evaluation. These identities do not assert a measured speedup. The spectral certificate needs $O(K)$ scalar terms once the angular parameters are known, but its useful cutoff grows as the buffer shrinks. Rounding, logarithms, trigonometric evaluation and the unknown stiffness need validated enclosures for a numerical certificate.

### 5.1 A worked two-arc example

Take the plus arc $(0,\pi/2)$, and enlarge it to $(0,\pi)$. Thus $\delta = \pi/2$. At $\rho = 1/2$, $T = \log 2$, $q = 1/4$, the constant and first two positive modes give exactly

$$E_{1/2,2} = \frac{\pi}{2\log 2} + \frac{32}{5\pi}, \qquad 0 \leq E_{1/2} - E_{1/2,2} \leq \frac{512}{567\pi}. \tag{5.7}$$

Indeed $1-\cos(\pi/2) = 1$, $1-\cos\pi = 2$, and the two terms in (5.2) are $16/(3\pi)$ and $16/(15\pi)$. The improved one-endpoint coefficient bound $|d_n| \leq 2/(\pi n)$ would divide the general tail in (5.1) by four; the displayed bound intentionally uses the two-arc certificate unchanged. Divide these energy endpoints by $32\pi^2 c_*$ to enclose the relative entropy. No numerical value of $c_*$ is inserted.

### 5.2 Global distinction and its limitation

For any different boundary data $B \neq 0$ in $L^2$, (5.5) makes the interior entropy tend to infinity and the interior total variation tend to one. The full distribution laws on $\mathbb{D}$ are in fact mutually singular. To see this without assuming a boundary trace of the random field, choose radii with normal-testing error at most $2^{-j}$ in (4.1). The likelihood threshold sets have summable wrong-decision probabilities under the two full laws. Their eventual-decision event has probability one under one law and zero under the other, by the first Borel–Cantelli lemma. Local equivalence therefore does not extend to an unbuffered global equivalence.

## 6 The existing normalization series and a conditional certificate

The series in this section belongs to the source; it is not a result newly discovered here. Let $A_{\mathbb{T}} = \sqrt{3}/2$, and let $q_0$ be the source's explicitly chosen even smooth cutoff, equal to one on $[-1/2, 1/2]$ and zero outside $(-1, 1)$. Define on the reciprocal torus

$$F_j(p) = q_0(4|p|)\cos(4\arg p)[q_0(2^j p_1) - q_0(2^{j+1}p_1)], \tag{6.1}$$

$$d_{ju} = \frac{A_{\mathbb{T}}}{(2\pi)^2}\int_{\widehat{\mathbb{T}}} F_j(p)e^{-ip\cdot u}\,dp. \tag{6.2}$$

These $d_{ju}$ are unrelated to the boundary coefficients $d_n$. Write $V(u) = \mathbb{E}(H(u) - H(0))^2$ in the source plane law, and $b_j = \sum_u d_{ju}V(u)$. The source proves

$$\sum_{j\geq 0}|b_j| < \infty, \qquad \pi c_* = \sum_{j\geq 0} b_j. \tag{6.3}$$

Its finite value $b_{js}$ uses free coherent spin pairs on a hexagon of radius $R_{js} = 2^{(j+s+4)^2}$, retaining $|u| \leq \sqrt{R_{js}}$. Increments are evaluated using (1.2); path independence is essential. For some source constants $K, A, \beta > 0$,

$$|b_{js} - b_j| \leq K2^{Aj}R_{js}^{-\beta}. \tag{6.4}$$

The source double series telescopes to (6.3) and is absolutely convergent. The coefficients involve prescribed Fourier integrals and ratios of finite integer counts, not only rational numbers.

**Proposition 6.1** (Separated finite and angular errors)**.** *Suppose certified constants in (6.4) are available, as well as an upper budget $T_J \geq \sum_{j\geq J}|b_j|$. For each $0 \leq j < J$, choose a radius index $s_j$, and a computed enclosure $|\widetilde{b}_j - b_{j,s_j}| \leq e_j$ including Fourier quadrature and count evaluation. Put*

$$\begin{aligned} S &= \sum_{j<J}\widetilde{b}_j, \qquad B_J = T_J + \sum_{j<J}\left(e_j + K2^{Aj}R_{j,s_j}^{-\beta}\right), \\ c_- &= \max\{0, (S - B_J)/\pi\}, \qquad c_+ = (S + B_J)/\pi. \end{aligned} \tag{6.5}$$

*Then $c_* \in [c_-, c_+]$, and $4\pi\sqrt{c_-} \leq \sigma \leq 4\pi\sqrt{c_+}$. If $c_- > 0$ and $E_\rho \in [E_-, E_+]$, then*

$$\frac{E_-}{32\pi^2 c_+} \leq \mathrm{KL}(\mu_J^\rho \| \mu_I^\rho) \leq \frac{E_+}{32\pi^2 c_-}. \tag{6.6}$$

*The analogous total-variation endpoints follow from (4.1).*

*Proof.* Subtract the finite sum from (6.3). The omitted angular sum, retained finite-volume errors and arithmetic errors give $B_J$ by the triangle inequality. The square root is monotone on nonnegative numbers. Finally $\sigma^2 = 16\pi^2 c_*$ and (4.1) give (6.6). The source guarantees positivity, but a certificate with $c_- = 0$ supplies no finite upper entropy bound. □

If the same source proof supplies (6.4) at arbitrary sufficiently large radii, explicit constants allow replacing the quadratic log-radius schedule by a linear one:

$$R_{js} \geq R_0 \quad \text{and} \quad \log_2 R_{js} \geq \frac{(A+\lambda)j + \mu s + B}{\beta}, \qquad \lambda, \mu > 0.$$

Then the error is at most $K2^{-B-\lambda j-\mu s}$, whose double sum is at most $K2^{-B}/[(1-2^{-\lambda})(1-2^{-\mu})]$. The telescoping proof of absolute convergence is unchanged. This is a conditional radius-budget

refinement, not an executed counting algorithm or an effective replacement when $A, \beta, K, R_0$ are unknown.

Absolute convergence alone gives no numerical $T_J$. An arbitrary finite prefix of an absolutely summable sequence can be followed by a single term of arbitrarily large magnitude. The source's angular tail depends on its dispersion-defect and alias measures; it is not certified by a finite count prefix. No numerical $c_*$, counting runtime, or explicit lattice convergence rate is reported here.

### 6.1 Why no extra triangular-area factor appears

The unit nodal hat has integral $A_{\mathbb{T}}$. At mesh $\delta$, the physical coefficient is $\delta^2 \int \lambda_0(y) f(\delta u + \delta y)\, dy$, with exact total $\int f$. Its cell mass cancels the site density $(A_{\mathbb{T}}\delta^2)^{-1}$ in the continuum transform. Thus the plane covariance is $c_* \int |\widehat{f}(\xi)|^2 |\xi|^{-2}\, d\xi$, whereas the declared inverse-Laplacian covariance has multiplier $(2\pi)^{-2}|\xi|^{-2}$. This gives $v = 4\pi^2 c_*$. The odd height fluctuation is twice the integer one, so $\sigma^2 = 4v$. The reciprocal factor in (6.2) has already been used. Inserting a second cell-area factor would be an error, not a coordinate convention.

## 7 Source boundaries, literature and consequences

### 7.1 Conditional law is stronger than geometric separation

The source's Gaussian collision proof fixes an exterior stopping record that includes every absolute spectator height. Its plane copy has the same conditional plane marginal for *each* record, making it independent of that record. A copy with only the correct unconditional marginal would not justify factorization. Equality of increments only removes a common offset on zero-total tests. Absolute tests require the source's equal-offset projection and deterministic boundary references. Similarly, the source reflection-null estimate is a null *reflection norm*, not vanishing ordinary variance. The local results of this report replace none of these steps.

Moment harmonicity holds first away from collisions. The source then uses an $O(r^2)$ collision-tube volume and a uniform polylogarithmic mollified moment bound, together with exact harmonic equality off that tube, to exclude hidden diagonal distributions. Only after logarithmic singularities and boundary means have been identified does harmonic uniqueness give the Gaussian recursion. A finite positive covariance matrix does not establish these distributional facts or tightness.

### 7.2 Primary public context as checked on October 9, 2026

The primary records for Glazman–Manolescu [2] (version 3, February 27, 2023) and Glazman–Lammers [3] (version 3, October 27, 2023) were checked afresh. They establish logarithmic fluctuations and related loop/delocalization results; the latter abstract describes these as consistent with a conjectural GFF limit. The March 6, 2026 primary six-vertex preprint [4] concerns a square-lattice full-plane GFF limit in its specified parameter range, not this triangular two-arc law. These external checks were abstract/metadata checks, not complete audits of their proofs. The present cached source is a separate September 25, 2026 manuscript, and its claimed resolution is attributed to that source. The search is not a proof that no other relevant work exists.

The Gaussian Hilbert-space and conformal-energy conventions were checked in Sheffield's primary text [5], especially its definition of the energy space, covariance pairing and orthogonal decomposition. The likelihood calculation above is proved explicitly. The minimum-energy principle, Fourier separation, Poisson kernel, Gaussian exponential tilt and Basel sum are classical; no historical priority is asserted for them. The private original Ram sources contribute domain and

physical-energy discipline, not a new law for Lipschitz heights. Their contents were not sent to public search services.

### 7.3 Advantages, costs and extensions

The boundary certificate is exact at every fixed $\rho < 1$, keeps the source's unknown normalization visible, and supplies a full complement bound after finitely many angular modes. The constant mode is included; dropping it would erase genuine information about unequal arc lengths. Conformal transport handles general smooth simply connected domains using a declared conformal observation region rather than a falsely universal Euclidean distance constant.

The costs are equally explicit. Computing a conformal map with certified boundary angles is an additional numerical problem. The Fourier tail deteriorates as $\rho \uparrow 1$. The required spectral tail budget for $c_*$ is not numerically provided by the source's absolute convergence. No inequality here gives a lattice rate or a uniform statement for marks coalescing with the mesh. The exact continuum small-motion coefficient can guide error budgeting, but it cannot be substituted for such a rate.

The formulas already extend to arbitrary $L^2$ boundary differences, including finitely many arcs; for bounded-variation data the coefficient bound $|d_n| \le \mathrm{Var}(B)/(2\pi n)$ gives the same style of tail enclosure. This is a continuum extension only: the source's two-arc pin and boundary proof has not been generalized to a multi-arc lattice theorem. Future work could supply verified angular-tail bounds for the normalization, validated conformal maps, or a genuine discrete relative-information estimate respecting the exact conditional law.

## 8 Verification and contribution boundary

The owned read ledger records every cached public file, with literal display intervals, before/after hashes and unread complements. All ten scientific sections were read in bounded displays, along with the main file, bibliography and two figure-source files. Imported external proofs were not thereby read. Fresh private reads covered the original TR8 elliptic/Green material and original TR250 variational/flux proofs; unread portions remain explicitly excluded. One truncated locator search was used only for navigation, not as a literal proof-read claim.

The owned exact-check script passed 766 parameter cases and 7,087 explicit assertions using exact rational arithmetic. It checks annular mode values, flux energies and positive tails, and separately checks the algebraic normalization and the worked example. Finite fixtures supplement the infinite-dimensional proofs; they neither simulate Lipschitz heights nor certify the source's global theorem. Final case/assertion counts, source hashes, native-compiler outcome and every-page visual receipt are recorded in the owned audit files.

The assessed increment is the buffered two-arc comparison, its two positive representations and sharp order-sensitive regimes, with a conditional stiffness enclosure. Applicability remains L3. C3 denotes this notable local analytical extension, not a global field proof, historical novelty, a new exponential-tail theorem, or a repeated credit for M443.

## References

[1] OpenAI, *The Gaussian free field limit of integer Lipschitz heights with two-arc boundary data*, September 25, 2026. Pinned public manuscript. Local scientific locators: Sections 02–10; captured source identity is provenance only.

[2] A. Glazman and I. Manolescu, *Uniform Lipschitz functions on the triangular lattice have logarithmic variations*, Commun. Math. Phys. 381 (2021), 1153–1221. Primary revised version.

[3] A. Glazman and P. Lammers, *Delocalisation and continuity in 2D: loop O(2), six-vertex, and random-cluster models*, Commun. Math. Phys. 406 (2025), article 108. Primary preprint record.

[4] H. Duminil-Copin, K. K. Kozlowski, P. Lammers and I. Manolescu, *Gaussian free field convergence of the six-vertex model with* $-1 \leq \Delta \leq -1/2$, 2026. Primary preprint.

[5] S. Sheffield, *Gaussian free fields for mathematicians*, Probab. Theory Relat. Fields 139 (2007), 521–541. Primary text, version 3; energy conventions and Gaussian Hilbert spaces, Sections 2.1, 2.4 and 2.6.

[6] M. Subbarao, original private TR8, local `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`. Fresh scoped reads: lines 65–197, including coercive inverse and boundary-aware Green equivalence. Confidential local evidence, not a public link.

[7] M. Subbarao, original private TR250, local `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Fresh scoped reads: lines 341–420 and 481–594, including variational convergence and the classical hypercircle/source-discrepancy proofs.